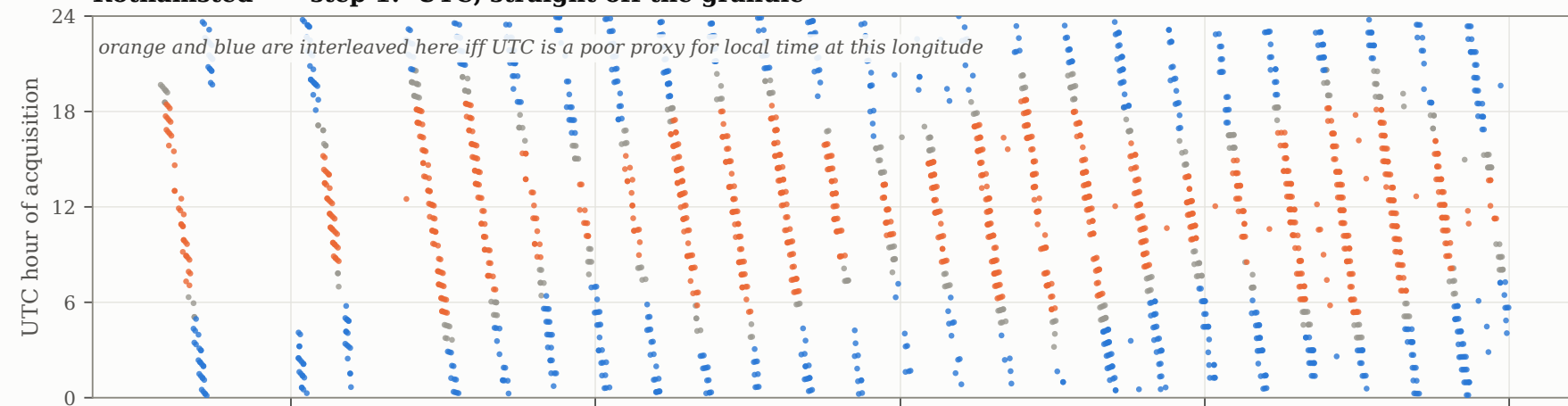
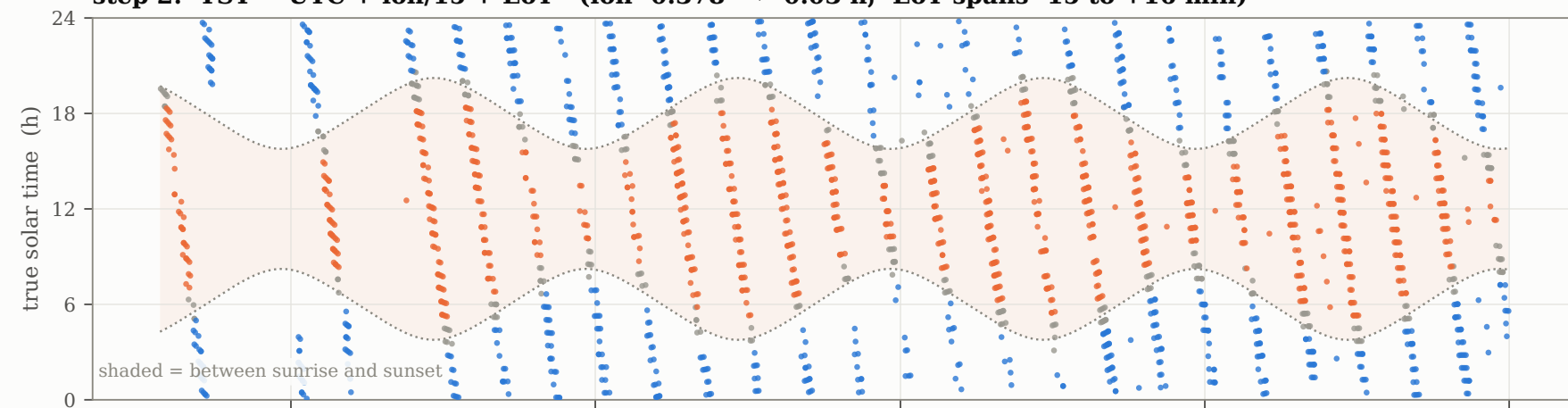


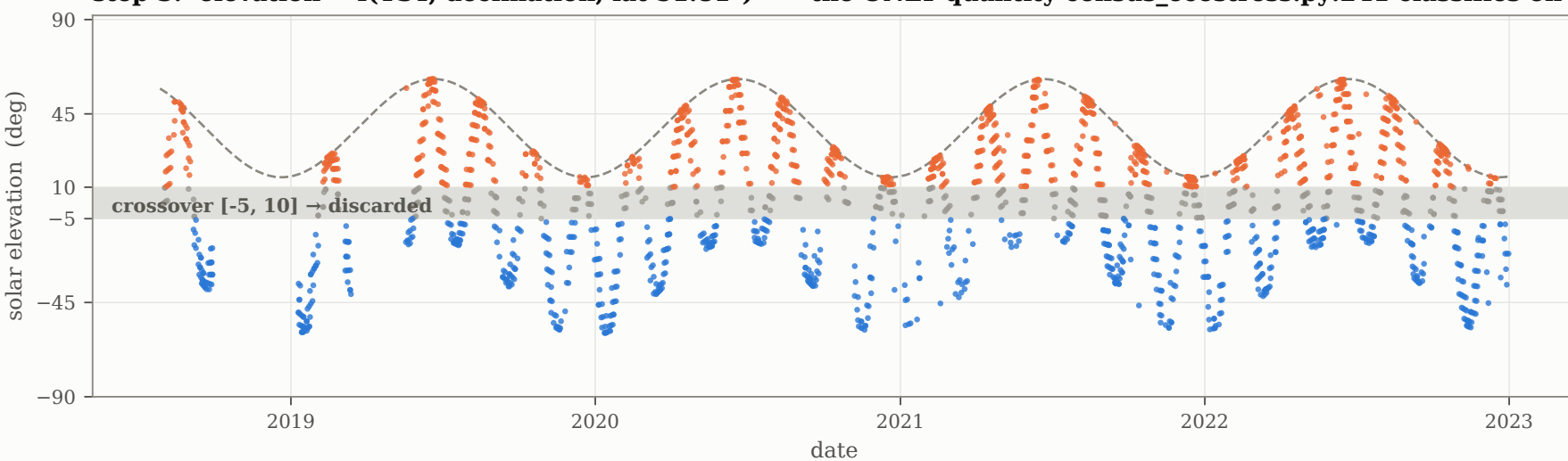
Rothamsted — step 1: UTC, straight off the granule



step 2: TST = UTC + lon/15 + EoT (lon -0.378° → -0.03 h; EoT spans -15 to +16 min)



step 3: elevation = f(TST, declination, lat 51.81°) — the ONLY quantity census_ecostress.py:241 classifies on



worked examples

UTC	TST	elev	phase
2018-08-03T17:42	17.58	17.4	day
2019-03-08T04:09	3.93	-23.2	night
2019-08-26T08:28	8.43	29.8	day
2020-02-16T10:29	10.22	21.2	day
2020-06-23T10:08	10.08	54.1	day
2020-11-07T04:14	4.48	-27.7	night
2021-04-24T08:36	8.62	33.6	day
2021-09-06T03:27	3.46	-17.4	night
2022-01-02T05:15	5.17	-25.4	night
2022-05-11T01:22	1.41	-18.0	night
2022-08-19T10:47	10.71	47.6	day
2022-12-31T05:40	5.58	-21.7	night

longitude offset -0.03 h
median |TST - UTC| 0.10 h

UTC is nearly useless to convert here — this station sits on the prime meridian, so UTC and solar time already agree.